

Sustainability Energy Partnership of Southeast PA (SEP)

Energy Cost Management and Procurement Plan

June 2023 (v.1)

I. Plan Statement

The goal of the SEP Energy Cost Management and Procurement Plan (ECMPP) is to mitigate and stabilize expected electric supply costs over a rolling four-year time horizon while providing the flexibility to incorporate renewable energy purchases to meet SEP member agency sustainability goals. No later than 6 months prior to the beginning of the upcoming fiscal year, electric supply costs will be stabilized within a predetermined limited variance range and the successive three fiscal years' costs will gradually be stabilized within a predetermined limited variance range over the next three years. The Plan will cover 4 years beyond the current fiscal year however it may be modified in the future to accommodate longer term financial commitments for renewable energy purchases. Market price targets may be considered to provide opportunities to reduce costs in future fiscal years relative to the current fiscal year.

II. Electric Account Profile

The current approximate annual purchase volumes behind the local utilities (PECO-349 acts, Met ED-19 acts, PPL-15 acts.) for the 4 participating member agencies are:

Approximate # Accounts	Calendar Yr. 2020 Annual (MWhs)
383	97,300

III. Electric Component Price Risk Profile

The electric supply component costs based on last 5- year average costs (AEPS costs based on last three years) and their respective price volatility profiles are as follows (excludes PA Gross Receipts Tax):

Supply	\$/MWh	% of Total	Price Volatility Profile
Energy	\$37.00	74%	Medium to High
Capacity	6.00	12%	High year to year
Transmission	2.50	5%	Low to Medium
PA AEPS & other	4.50	9%	Medium

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Energy

Over the last five years approximately **74%** of the costs represent energy costs that are determined directly by energy commodity/trading markets. While energy prices were relatively low and stable from spring of 2016 thru spring of 2021 (5 years), prices thereafter became significantly more volatile driven by worldwide economic and political developments and more extreme weather events. Forward and spot (hourly) electric prices in PJM, (the mid-Atlantic regional grid that includes the PECO service territory) are driven primarily by natural gas prices and the historical and current correlation between electric and natural gas prices is over 90%. Given the current PJM market structure/design, this high correlation is expected to continue for the foreseeable future. Longer term, if significant renewable energy capacity is developed in the PJM region, the correlation with natural gas prices could decline (as experienced with solar in California). Below is a summary of the electric forward and spot market volatility over the last 5 years (June 2018-May 2023) for the PECO Zone.

Energy Cost Volatility Range Summary

Markets	Low		High	
	Price (\$/MWh)	SEP Cost (1) (\$ millions)	Price (\$/MWh)	SEP Cost (1) (\$ millions)
Hourly (2)	\$19.29	\$1.88	\$49.30	\$4.80
Forward (3)	\$21.09	\$2.05	\$108.17	\$10.52

- (1) Based on **97,300** MWhs per annum
- (2) Based on PJM monthly average day ahead around the clock (ATC) hourly prices at PECO delivery zone for the period June 2018 through May 2023.
- (3) Based on ATC forward market prices for a one year delivery period at the PECO delivery zone.

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Capacity

Represents approximately 12 % of the total cost and this amount will vary each year based on PJM capacity clearing prices and the portfolio's annual peak load contribution (PLC). The capacity prices and PLCs are based on PJM's reporting year June 1 - May 31. Historically capacity prices have been published three years in advance as the auction is held each year for the next third year out, however ongoing changes to the PJM capacity market design has caused prices to be determined less than 2 years in advance. The annual auction clearing prices are determined based on the projected demand and available generation capacity to serve that demand (supply) by utility zone.

The capacity prices and cost range for the PECO zone for the most recent published five (5) years (20/21 - 24/25) are:

Capacity Price (\$/MWD)

\$50.96 - \$174.32

Estimated Annual Capacity Cost (\$)*

\$279,000 - \$954,000

*Based on estimated SEP unforced capacity obligation (UCAP) of 15.0 MW. UCAP represents the PLC grossed up for PJM reserve and other system adjustments

MWD = MegaWatt Day

Annual capacity costs can be controlled/reduced by reducing demands during summer peak hours (peaks hours generally occur during July and August, Monday-Friday, hours ending 4:00 P.M. - 6:00 p.m. EST). Enel X will provide SEP member agencies with a system peak predictor service to assist in peak load reduction management.

Transmission Costs

Transmission costs (5% of total supply costs) (referred to as network integrated transmission services (NITs)) are passed thru by PJM based on each account's peak load incurred during PECO's service territory highest peak hour occurring during the prior November 1 – October 31 12 month period. The Transmission obligation (KW) for each account is reset on January 1 of each year following the October 31 peak load end date. NITs rates expressed in \$/MWY (MegaWatt Year) can change at any time based on PECO's rate filing with PJM/FERC however NITs rate changes usually occur effective Jan 1 or June 1.

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PA AEPS & Other Costs

The remaining portion of supply costs (approximately 9%) consist primarily of Pennsylvania Alternative Energy Portfolio Standard (AEPS) costs which are also referred to as regulatory required renewable energy costs. Under AEPS, designed percentages of the energy purchased must be derived from qualifying Tier I, Tier II and solar resources. This is achieved by purchasing Tier I, Tier II and PA Solar renewable energy certificates (RECs) which are traded/purchased in the marketplace. With the increasing renewable energy requirements in adjacent states, AEPS compliant RECs prices have been increasing and absent changes in the AEPS requirements brought about by PA Legislation, could continue to increase over time.

IV. Procurement Strategy and Purchase Structure

SEP utilizes a managed portfolio strategy for managing energy supply costs. Under this approach all component cost items are passed thru at cost with energy market exposure managed on an active ongoing basis. The purchase strategy is executed through a dedicated PJM subaccount competitively procured from a Pennsylvania licensed service provider. All SEP accounts are enrolled into the subaccount by the service provider. The energy costs for all metered consumption reported in the subaccount are based on a combination of fixed rate wholesale block purchases and PJM PECO zone hourly prices in effect during the reporting period. All other PJM charges (capacity, transmission, ancillary costs) associated with the accounts are passed thru to the subaccount at cost. The Licensed Services Provider (LSP) bills the accounts based on pre-determined budgeted rates prepared by the Portfolio Energy Advisor (PEA) and approved by the SEP Energy Board. Billing rates are adjusted periodically to true up the supply portfolio's billed vs. actual costs.

V. Energy Price Risk Management Strategy

1. Flexible Dollar Cost Average Mechanism

The overall strategy is to build price stability by avoiding single point market exposure, i.e., making a fixed price commitment for 100% of energy requirements at a single point in time as 74% of the overall supply cost is determined in the commodity markets which can be volatile, especially during extreme weather events. This is to be accomplished through a dollar cost averaging mechanism where fixed rate blocks are purchased for portions of consumption over time such that by six (6) months prior to the beginning of a fiscal year the maximum desired level of price stability has been established for that period.

The level of fixed price commitments will increase for each period as it draws closer to the full commitment date (six months prior to fiscal year). SEP retains the flexibility to accelerate fixed price block purchases when market opportunities arise and defer block purchases when market prices are deemed unattractive or inflated.

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2. Implementation Guideline

Fixed Price Positions

The targeted hedged (fixed price) positions for the energy exposure of the entire portfolio will be:

80% - 90% - on peak expected volume

70% - 80% - off peak expected volume

The hedged positions will be targeted for execution six months prior to the next budget year.

Planned Hedge Position Timeline

	On Peak Exposure		Off Peak Exposure	
	% of Target Hedge	Accumulated % Target Hedge	% of Target Hedge	Accumulated % Target Hedge
27 Months prior to budget period:	25%	25%	25%	25%
18 Months prior to budget period:	25%	50%	25%	50%
12 Months prior to budget period:	25%	75%	25%	75%
6 Months prior to budget period:	25%	100%	25%	100%

VI. Renewable Energy

The portfolio will comply with the Pennsylvania Alternative Energy Portfolio Standard (AEPS), a regulatory requirement that certain portions of energy purchased is to include energy produced from renewable sources as defined in the AEPS regulation. The portfolio will incur higher compliance fees should it not meet the AEPS requirement. The AEPS requirement will be met through the purchase of qualifying renewable energy certificates (REC's) which the Energy Consultant will coordinate with SEP and the licensed electric supply services provider.

SEP is in the process of developing a renewable energy procurement strategy that will consider longer term renewable energy purchase commitments. Once the strategy has been developed and approved, it will be linked to this document.

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VII. Cost Management

The following cost management strategies and tactics will be employed to manage portfolio supply costs:

- SEP will purchase fixed rate wholesale blocks of power to manage energy price volatility. To help ensure competitive pricing, the LSP will obtain a minimum of three counterparty wholesale bids. The PEA will also provide a comparison of price offers to available published market prices to ensure price competitiveness.
- Block pricing may be obtained at the PJM West Hub and or PECO delivery zone and pricing will be selected based on pricing differentials and assessments of ongoing PECO congestion risk.
- Allow assumption of a defined amount of spot market risk to avoid liquidating fixed rate purchases against the spot market.
- Utilize Day Ahead Scheduling to reduce operating reserve costs and reduce spot market energy price volatility.
- Provide system peak predictor service to all member agencies to assist SEP members in reducing capacity costs and possibly transmission costs.
- Participate in annual PJM Auction Revenue Rights (ARR) process to influence level of ARR credits.
- Purchase qualifying PA AEPS REC's (Renewable Energy Certificates) on a monthly basis to avoid unpredictability of REC prices given overall lack of market liquidity and regulatory uncertainty. Consideration will also be given to term purchases as a hedge against escalating REC prices.

VIII. Program Management

Refer to the Program Operating Procedures (POP).

IX. Transaction Authority

Refer to the Program Operating Procedures (POP).